CLINICAL · MOUTH & THROAT

Applied anatomy and physiology of the pharynx

The pharynx is a shared corridor: air passes through it to the larynx and food through it to the oesophagus, and many times a minute it switches between the two without error. Its walls hold the lymphoid ring that guards both portals, its spaces carry infection down into the chest, and its three parts each give rise to their own cancers. The anatomy explains the symptoms: the earache of a tonsil cancer, the nasal voice of a palatal palsy, the pouch at Killian's dehiscence.

AT A GLANCE

- Fossa of Rosenmuller: commonest site of nasopharyngeal carcinoma.
- Killian's dehiscence lies between thyropharyngeus and cricopharyngeus.
- All pharyngeal muscles are vagal except stylopharyngeus (IX) and tensor palati (V3).
- The piriform sinus drains richly to the neck.

1 CHAPTER ONE The pharynx in general

A fibromuscular funnel.

1.1 Extent

A conical fibromuscular tube 12 to 14 cm long, from the base of the skull (basisphenoid and basiocciput) to the lower border of the cricoid cartilage at the level of the sixth cervical vertebra, where it becomes the oesophagus. It is about 3.5 cm wide at the top and narrows to 1.5 cm at the pharyngo-oesophageal junction, the narrowest part of the gut after the appendix, which is why swallowed foreign bodies lodge there.

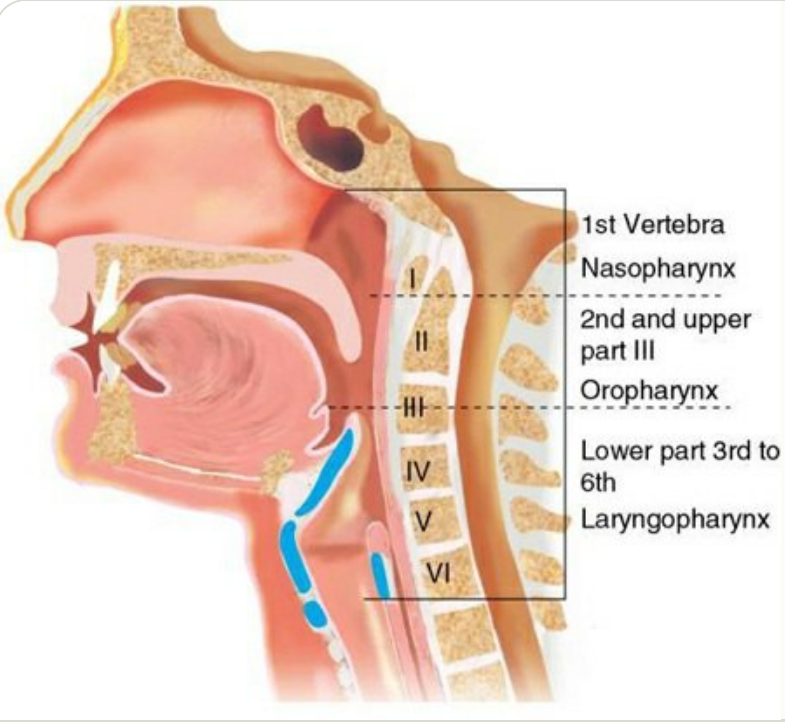
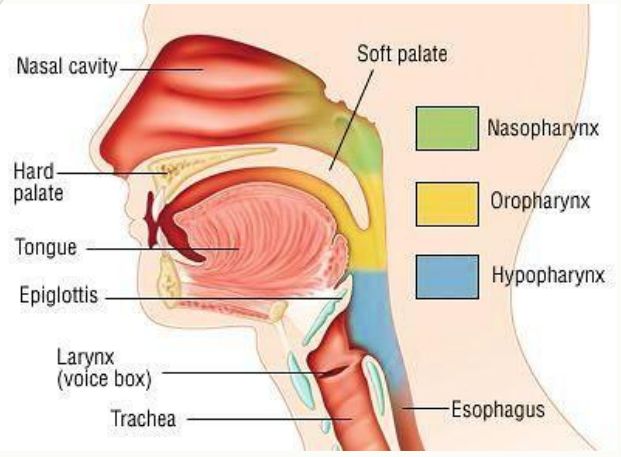


FIGURE 1 Divisions of the pharynx and the vertebrae related to the posterior wall: nasopharynx (C1), oropharynx (C2 and upper C3), and laryngopharynx (C3 to C6).

Dhingra, fig. 47.3



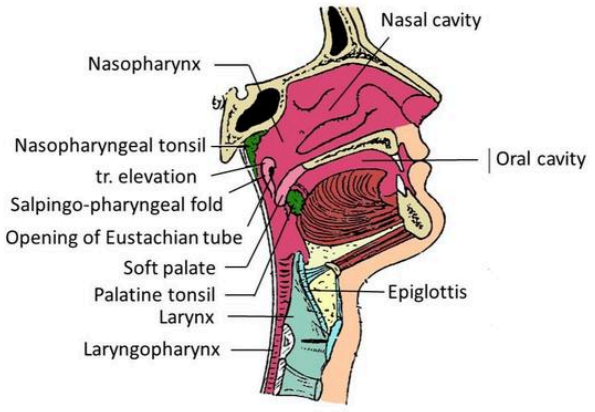


FIGURE 2 The divisions of the pharynx, and its cavity in sagittal section.

Anatomy atlas · ENT Steps p. 194, Doss p. 168

1.2 Layers of the wall

From within outwards:

- **Mucous membrane:** ciliated columnar in the nasopharynx, stratified squamous below; continuous with the lining of the Eustachian tubes, nose, mouth, larynx and oesophagus.

- **Pharyngobasilar fascia (pharyngeal aponeurosis):** thick at the skull base, where it fills the gap above the superior constrictor, thin below.
- **Muscular coat:** an outer circular layer (the three constrictors) and an inner longitudinal layer (stylopharyngeus, salpingopharyngeus, palatopharyngeus).
- **Buccopharyngeal fascia:** covers the constrictors and continues forwards over the buccinator.

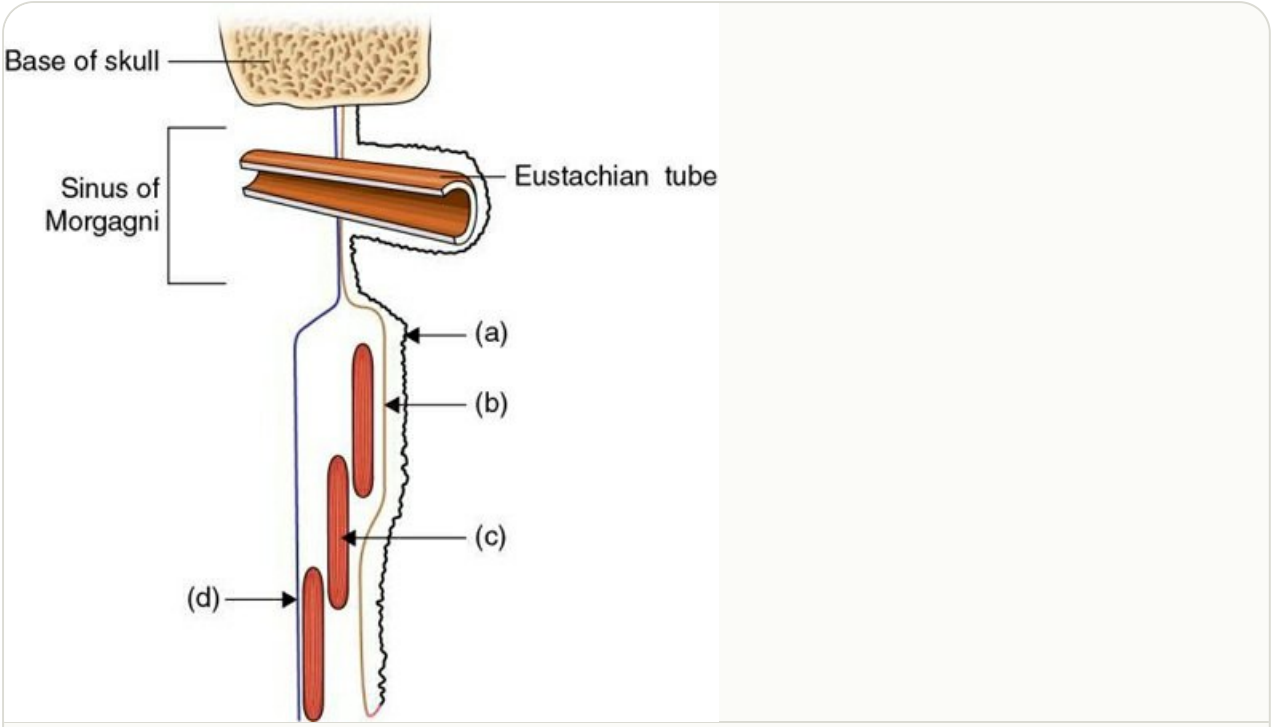


FIGURE 3 Structure of the pharyngeal wall: (a) mucous membrane, (b) pharyngobasilar fascia, © muscular coat, (d) buccopharyngeal fascia; the sinus of Morgagni lies above the superior constrictor.

Dhingra, fig. 47.1

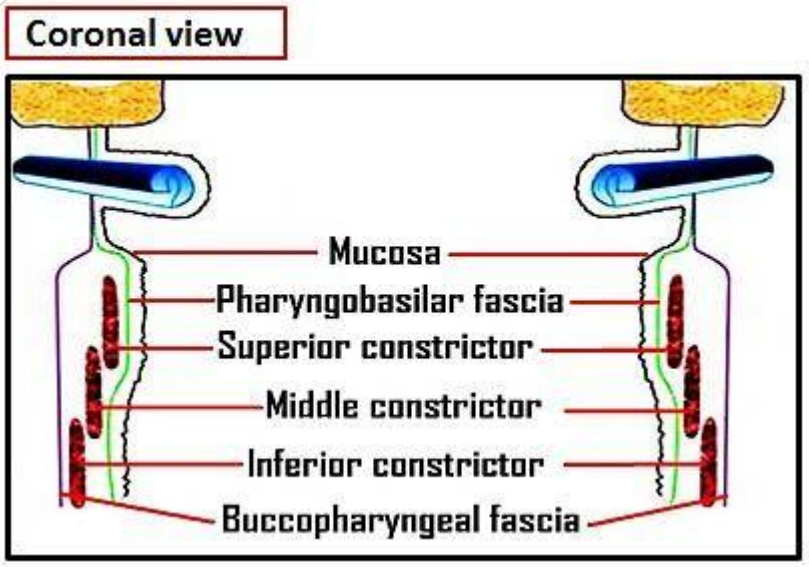


FIGURE 4 The layers of the pharyngeal wall.
Anatomy atlas · ENT Steps p. 197

CHAPTER TWO

2 Muscles and gaps

Three constrictors, three elevators.

2.1 Constrictors

Each passes backwards to insert into the midline raphe, the lower overlapping the upper like stacked cups. All are supplied by the pharyngeal plexus (vagus).

- **Superior:** from the pterygoid hamulus, pterygomandibular raphe and mandible.
- **Middle:** from the hyoid bone and stylohyoid ligament.
- **Inferior:** from the thyroid and cricoid cartilages, in two parts: the oblique *thyropharyngeus* and the transverse *cricopharyngeus*, which is the upper oesophageal sphincter, tonically closed at rest and relaxing on swallowing.

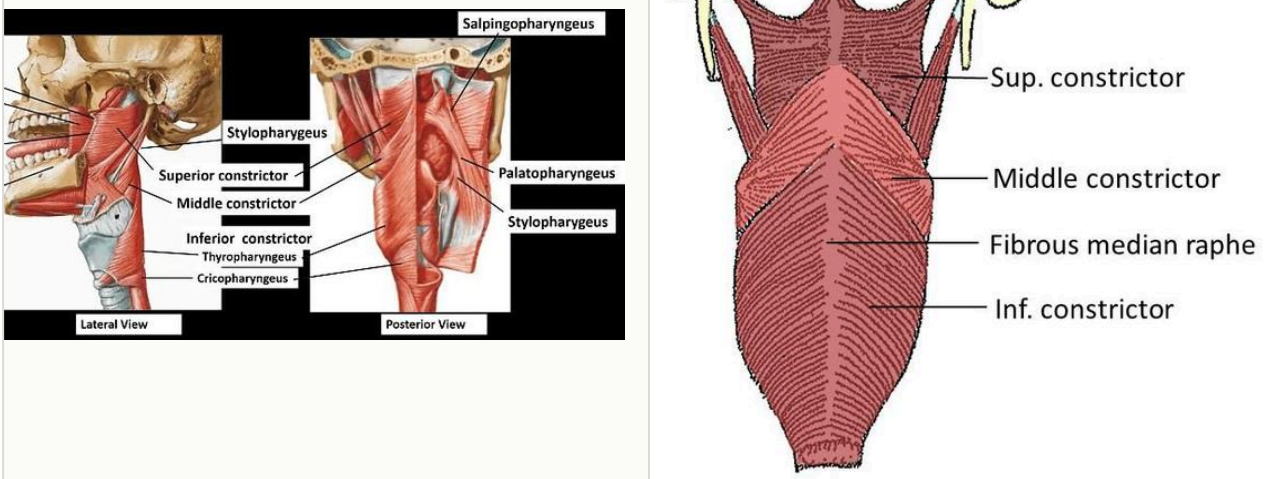


FIGURE 5 The pharyngeal constrictors in lateral and posterior views.
Anatomy atlas · ENT Steps p. 197, Doss p. 166

2.2 Longitudinal muscles

- **Stylopharyngeus:** from the styloid process; the only pharyngeal muscle supplied by the glossopharyngeal nerve.
- **Salpingopharyngeus:** from the Eustachian tube cartilage; raises the salpingopharyngeal fold.
- **Palatopharyngeus:** from the palate; forms the posterior pillar and, with fibres of the superior constrictor, *Passavant's ridge*.

All elevate the pharynx and larynx during swallowing.

2.3 Gaps and what passes through them

- **Above the superior constrictor (sinus of Morgagni):** the Eustachian tube, levator and tensor veli palatini, and the ascending palatine artery.
- **Between superior and middle constrictors:** stylopharyngeus and the glossopharyngeal nerve.
- **Between middle and inferior constrictors:** the internal laryngeal nerve and superior laryngeal vessels.
- **Below the inferior constrictor:** the recurrent laryngeal nerve and inferior laryngeal vessels.

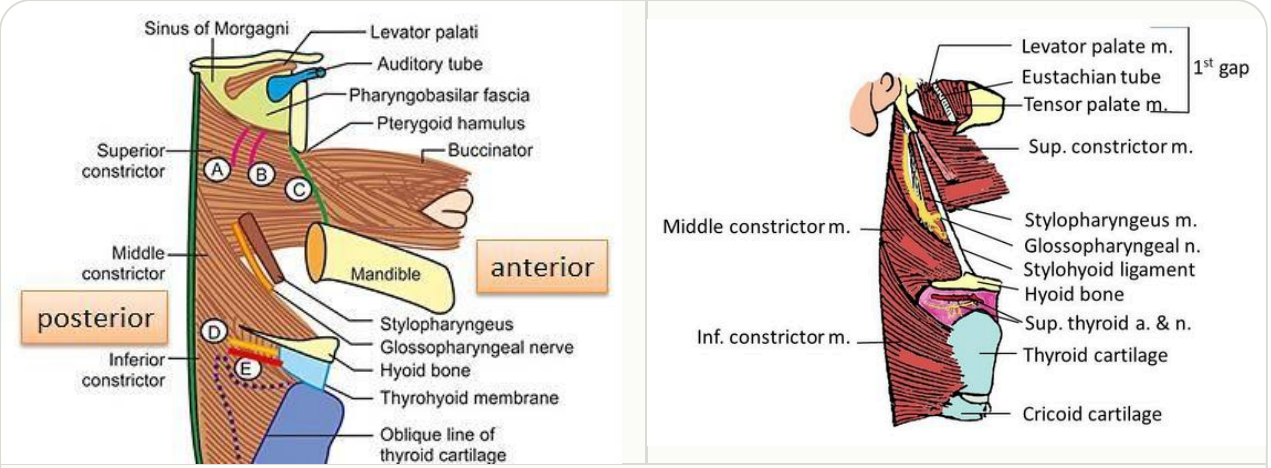


FIGURE 6 The gaps between the constrictors and the structures passing through them.
Anatomy atlas · ENT Steps p. 204, Doss p. 167

2.4 Killian's dehiscence

A weak triangle in the posterior wall between thyropharyngeus above and cricopharyngeus below. Mucosa herniates through it to form a *pharyngeal pouch* (Zenker diverticulum), and an oesophagoscope can perforate it, hence its old name, the *gateway of tears*. Below cricopharyngeus lie Laimer's triangle and the lateral Killian–Jamieson area.

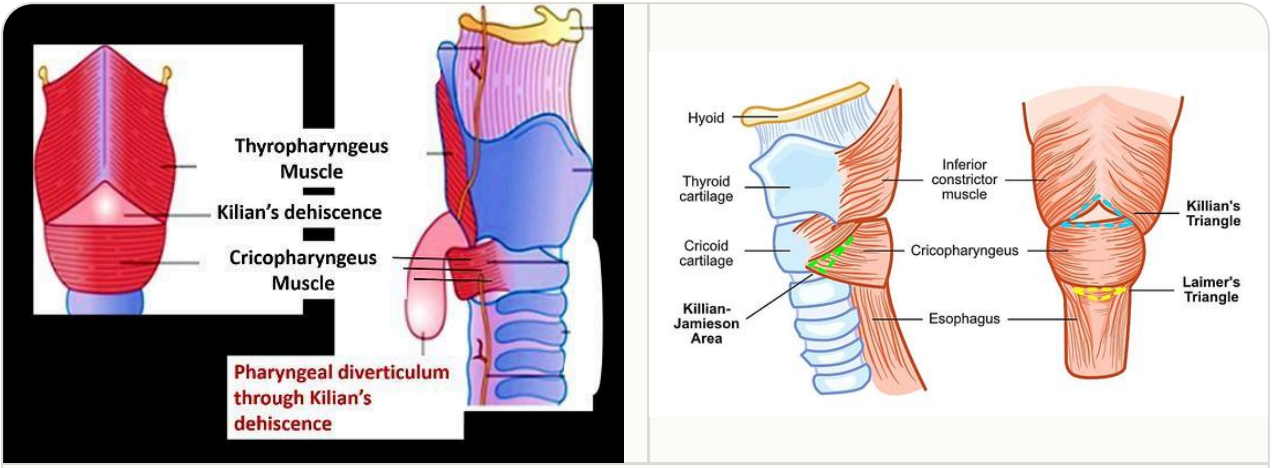


FIGURE 7 Killian's dehiscence and the pharyngeal pouch, and Killian's and Laimer's triangles.
Anatomy atlas · ENT Steps p. 198, ENT Steps p. 225

3 CHAPTER THREE Waldeyer's ring

Sentinels at the portals.

3.1 Components

Lymphoid tissue lies throughout the pharyngeal submucosa and gathers into masses that ring the entrances of the air and food passages:

- **Nasopharyngeal tonsil (adenoids)** at the junction of roof and posterior wall.
- **Palatine tonsils** between the pillars.
- **Lingual tonsil** on the base of tongue.
- **Tubal tonsils** around the Eustachian tube openings.
- **Lateral pharyngeal bands** behind the posterior pillars.
- **Nodules** scattered on the posterior wall.

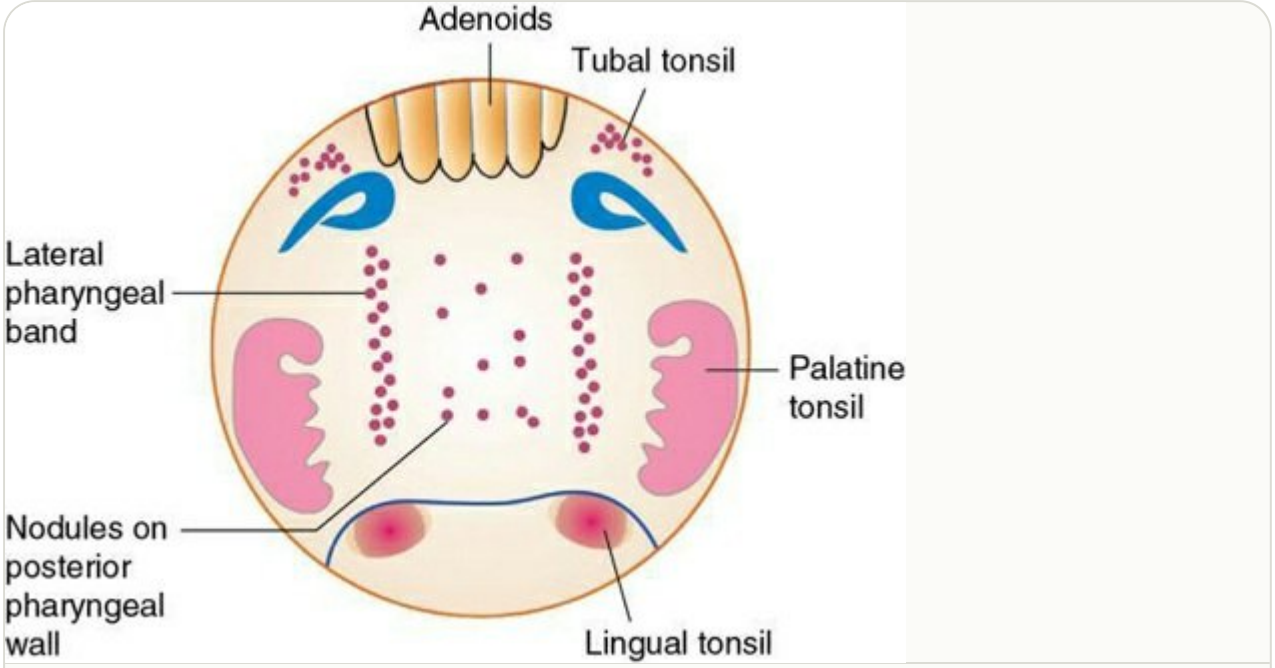


FIGURE 8 Waldeyer's ring.
Dhingra, fig. 47.2

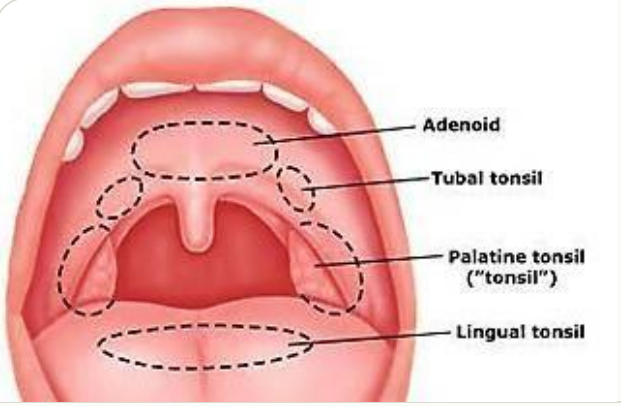


TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

FIGURE 9 Waldeyer's ring seen through the mouth, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 199, ENT Steps p. 208

3.2 Immunology

B-lymphocytes in the germinal centres produce secretory IgA, T-lymphocytes in the parafollicular areas give cell-mediated immunity, and plasma cells secrete IgM and IgG against organisms that enter the crypts. The tissue is most active in childhood. No lasting immune deficit has been shown after removal of the tonsils and adenoids.

4 CHAPTER FOUR Nasopharynx

Behind the nose.

4.1 Boundaries

Behind the nasal cavities, from the skull base to the level of the hard palate.

- **Roof and posterior wall:** basisphenoid and basiocciput, curving down onto the arch of the atlas.
- **Floor:** the upper surface of the soft palate in front; behind it the *nasopharyngeal isthmus*, closed by the palate in swallowing and speech.
- **Anterior wall:** the choanae and the posterior ends of the turbinates.
- **Lateral wall:** the Eustachian tube opening, about 1.25 cm behind the inferior turbinate, capped above and behind by the *torus tubarius*; behind the torus, the *fossa of*

Rosenmüller, the commonest site of nasopharyngeal carcinoma; and the salpingopharyngeal fold descending from it.

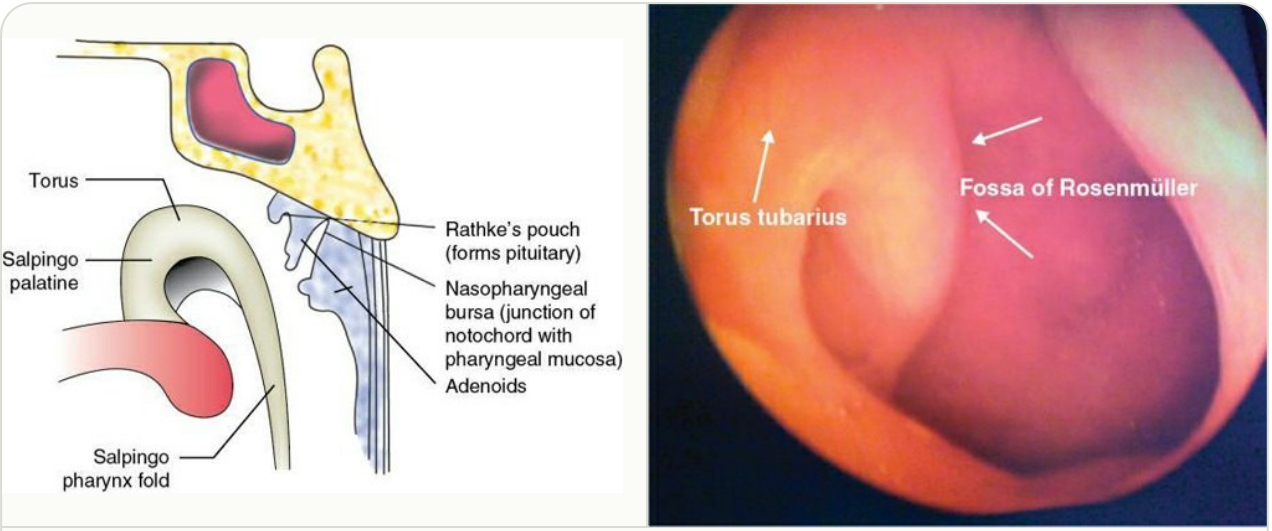


FIGURE 10 First, the nasopharynx in sagittal section with the adenoids, nasopharyngeal bursa and Rathke's pouch. Second, an endoscopic view of the torus tubarius and the fossa of Rosenmüller behind it.

Dhingra, figs 47.4 and 47.5

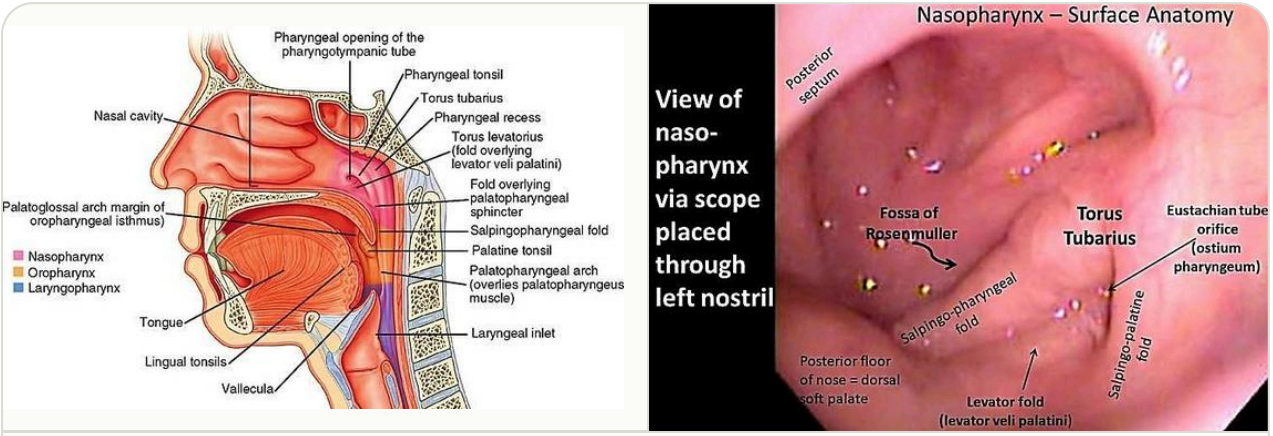


FIGURE 11 The lateral wall of the nasopharynx with the torus tubarius and fossa of Rosenmüller, and an endoscopic view.

Anatomy atlas · ENT Steps p. 203

4.2 Special features

- **Adenoids:** grow until about six years and then atrophy; the overlying mucosa is thrown into radiating folds.
- **Nasopharyngeal bursa:** a midline recess within the adenoid, marking where the notochord touched the pharyngeal endoderm; infection causes postnasal discharge, and an abscess or cyst is *Thornwaldt disease*.

- **Rathke's pouch:** a dimple above the adenoids, remnant of the invagination that forms the anterior pituitary; a craniopharyngioma can arise along its course.
- **Passavant's ridge:** a ridge around the isthmus against which the soft palate closes.

4.3 Lymphatic drainage

To the retropharyngeal (node of Rouvière) and parapharyngeal nodes, then the upper deep cervical and spinal accessory (posterior triangle) chains, often to both sides.

4.4 Functions

Air conduit; ventilation and pressure equalisation of the middle ear; velopharyngeal closure in swallowing, vomiting and speech; a resonator for the voice (hyponasal with obstruction, hypernasal with velopharyngeal insufficiency); and drainage of nasal mucus.

5 CHAPTER FIVE Oropharynx

Behind the mouth.

5.1 Boundaries

From the plane of the hard palate to the plane of the hyoid bone, opening forwards into the mouth through the oropharyngeal isthmus (soft palate above, tongue below, anterior pillars at the sides).

- **Anterior wall:** the base of tongue with the lingual tonsil, and the *valleculae*, each between the median and lateral glossoepiglottic folds, a site for retention cysts.
- **Lateral wall:** the palatine tonsil in its fossa between the anterior (palatoglossal) and posterior (palatopharyngeal) pillars.
- **Posterior wall:** in front of C2 and C3, with the retropharyngeal space behind.
- **Superior wall:** the undersurface of the soft palate.

The upper border of the epiglottis and the pharyngoepiglottic folds separate it from the hypopharynx.

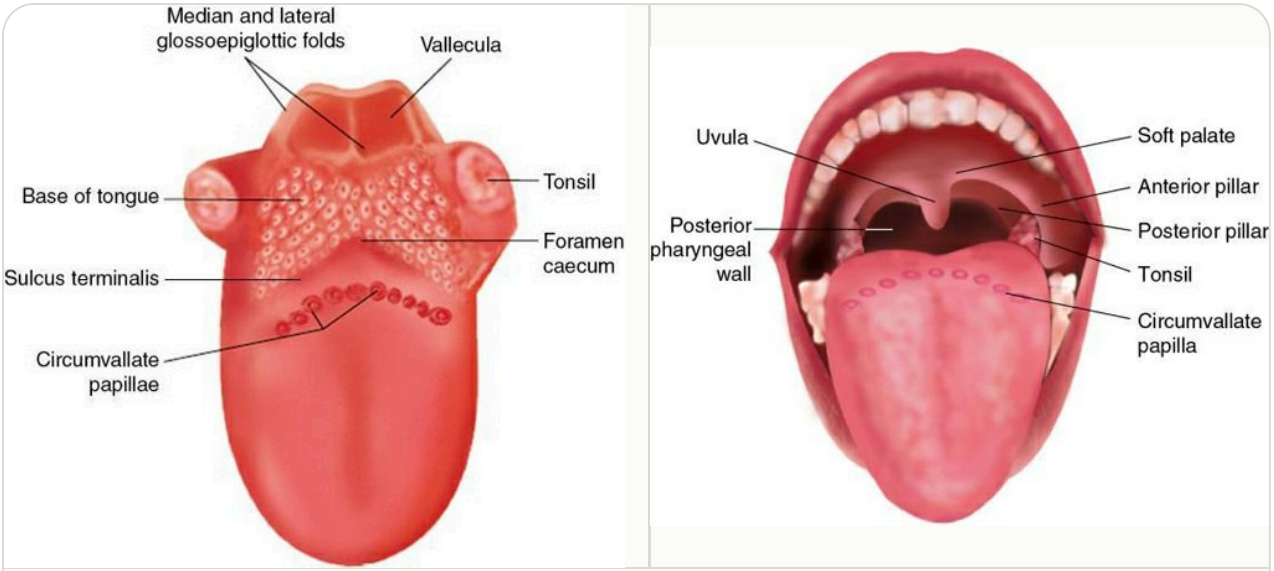


FIGURE 12 First, the base of tongue and valleculae. Second, the structures seen in the oropharynx: uvula, soft palate, anterior and posterior pillars, tonsil and posterior pharyngeal wall.
Dhingra, figs 47.6 and 47.7

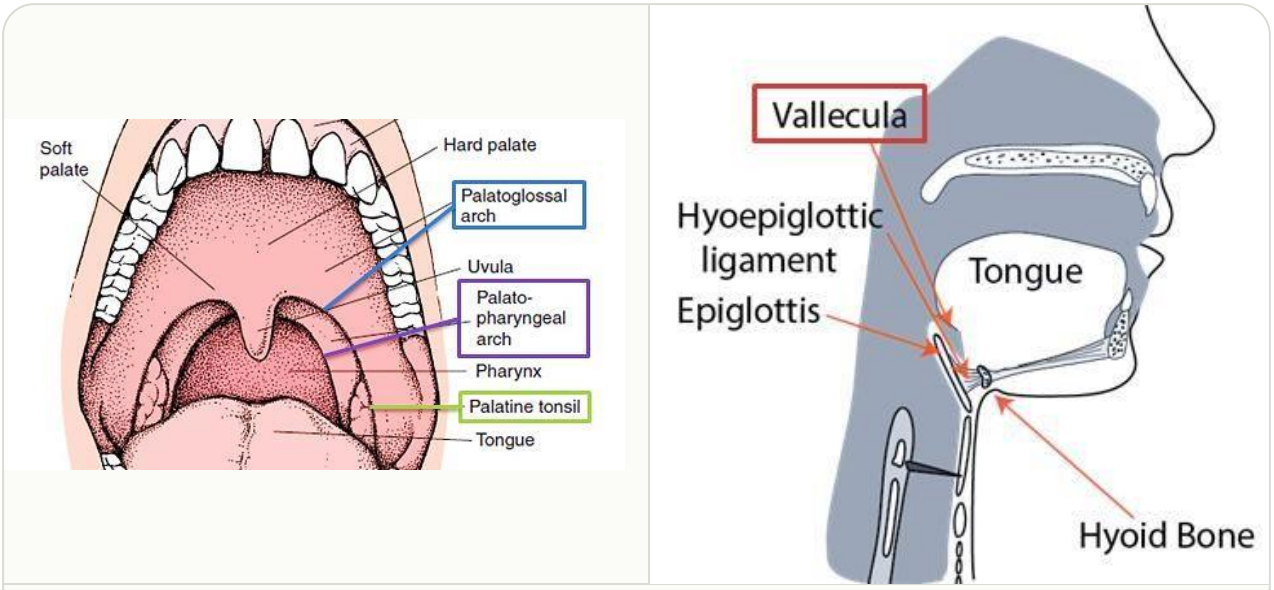


FIGURE 13 The oropharynx through the open mouth, and the vallecula with the hyoepiglottic ligament.
Anatomy atlas · ENT Steps p. 210

5.2 Soft palate

A mobile flap hung from the back of the hard palate. When it rises it seals the nasopharynx from the oropharynx; when it falls against the tongue it seals the

mouth, so we can chew and breathe together.

MUSCLES OF THE SOFT PALATE

MUSCLE	ACTION	NERVE
Levator veli palatini	Raises the palate	Vagus (pharyngeal plexus)
Tensor veli palatini	Tightens the palate; opens the Eustachian tube	Mandibular nerve (V3)
Palatoglossus	Narrows the isthmus; forms the anterior pillar	Vagus
Palatopharyngeus	Raises the pharynx; forms the posterior pillar	Vagus
Musculus uvulae	Shortens the uvula	Vagus

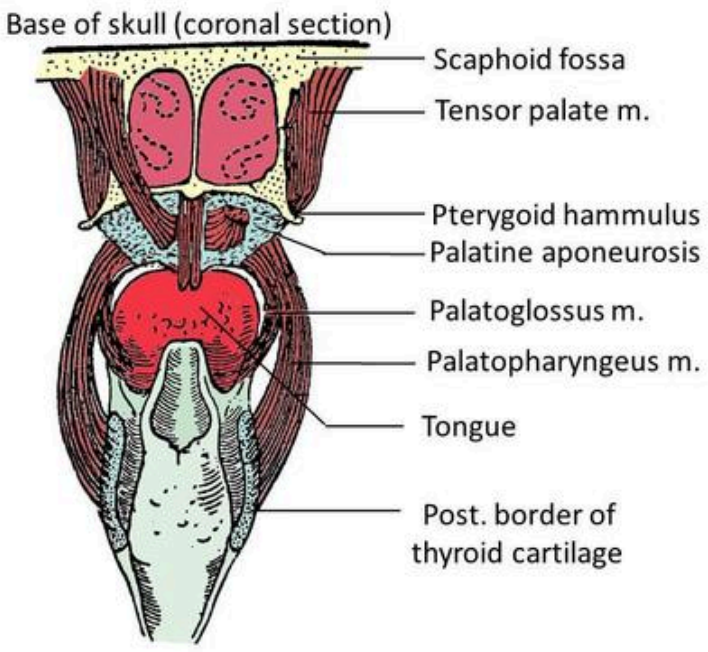


FIGURE 14 The muscles of the soft palate.
Anatomy atlas · Doss p. 181

THE UVULA IN A VAGAL PALSY

With a unilateral vagal (pharyngeal plexus) palsy the uvula and palate are pulled towards the healthy side on saying "aah". The tensor alone is supplied by the trigeminal nerve.

5.3 The palatine tonsil

Its anatomy, blood supply and bed are described with tonsillitis; it drains to the *jugulodigastric (tonsillar) node*.

5.4 Lymphatic drainage

To the upper jugular chain, especially the jugulodigastric node; the soft palate, pharyngeal walls and base of tongue also drain through the retropharyngeal and parapharyngeal nodes, and the base of tongue drains to both sides.

5.5 Functions

A passage for air and food, the pharyngeal phase of swallowing, shaping of speech sounds, taste (taste buds on the base of tongue, soft palate, pillars and posterior wall), and immune defence through Waldeyer's ring.

6 CHAPTER SIX Hypopharynx

Behind and beside the larynx.

6.1 Boundaries and subsites

From the plane of the hyoid bone to the lower border of the cricoid, in front of C3 to C6.

- **Piriform sinus (fossa):** on each side of the larynx, from the pharyngoepiglottic fold to the oesophagus; bounded laterally by the thyrohyoid membrane and thyroid cartilage, medially by the aryepiglottic fold, arytenoid and cricoid. It is the lateral food channel where fish bones lodge. The *internal laryngeal nerve* runs under its mucosa, so it can be anaesthetised there, and through it a piriform cancer refers pain to the ear.
- **Postcricoid region:** the anterior wall behind the cricoid lamina; the site of carcinoma in women with Plummer–Vinson syndrome.
- **Posterior pharyngeal wall:** from the hyoid to the level of the cricoarytenoid joints.

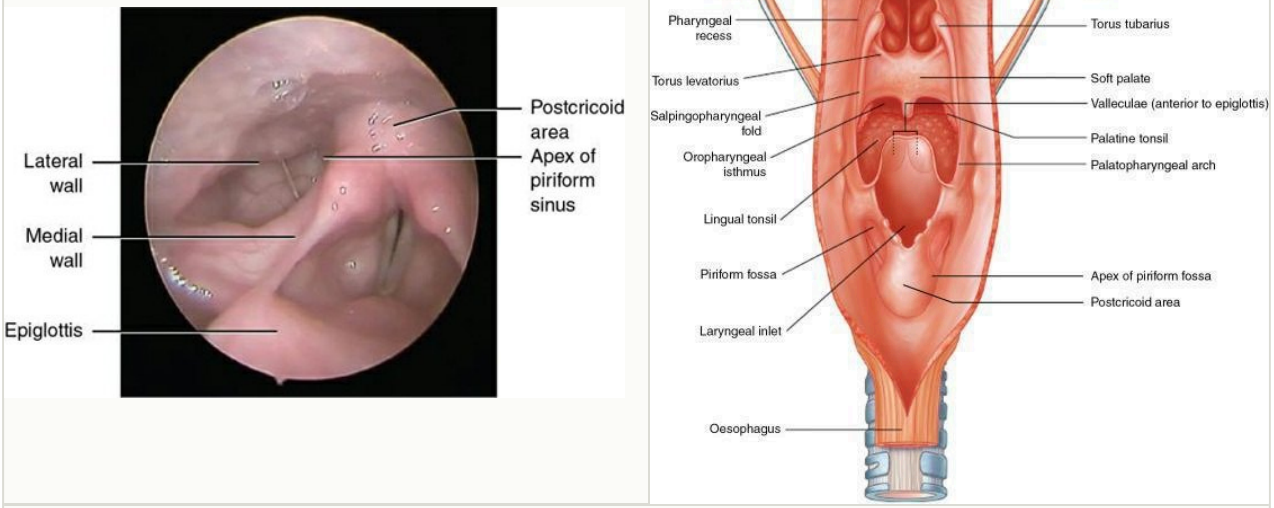


FIGURE 15 First, the right piriform fossa with its lateral and medial walls, apex and the postcricoid area. Second, the pharynx opened from behind, showing the nasopharynx, oropharynx and laryngopharynx.

Dhingra, figs 47.8 and 47.9

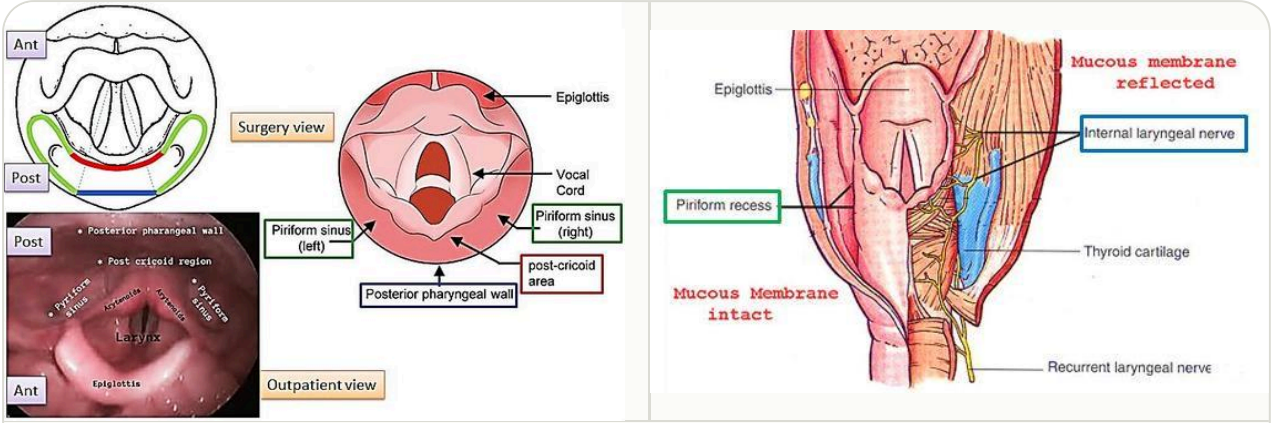


FIGURE 16 The three subsites of the hypopharynx, and the internal laryngeal nerve under the mucosa of the piriform fossa.

Anatomy atlas · ENT Steps p. 220, ENT Steps p. 221

6.2 Lymphatic drainage

The piriform sinus is richly supplied; its lymphatics pierce the thyrohyoid membrane to the upper jugular nodes, which is why its cancers so often present with neck nodes. The posterior wall drains to the retropharyngeal and parapharyngeal nodes; the postcricoid region to the parapharyngeal, paratracheal and supraclavicular nodes.

7 CHAPTER SEVEN Nerves, vessels and spaces

The supply lines.

7.1 Nerve supply

- **Motor:** the pharyngeal plexus (vagus) supplies all pharyngeal and palatal muscles except stylopharyngeus (glossopharyngeal) and tensor veli palatini (mandibular nerve).
- **Sensory:** nasopharynx by the maxillary nerve (V2); oropharynx by the glossopharyngeal nerve, the afferent limb of the gag reflex; hypopharynx by the internal laryngeal branch of the vagus.

REFERRED OTALGIA FROM THE THROAT

The glossopharyngeal nerve supplies the middle ear through Jacobson's nerve, and the vagus supplies the ear canal through Arnold's nerve. Tonsillitis, a quinsy, a post-tonsillectomy throat and cancers of the tonsil, base of tongue or piriform sinus all present with earache.

7.2 Blood supply

The ascending pharyngeal artery, the tonsillar and ascending palatine branches of the facial artery, branches of the maxillary and lingual arteries, and below, the inferior thyroid artery. Veins form a pharyngeal plexus draining to the pterygoid plexus and internal jugular vein.

7.3 Pharyngeal spaces

- **Retropharyngeal space:** behind the pharynx, in front of the prevertebral fascia, from the skull base to the bifurcation of the trachea; contains the retropharyngeal nodes, which regress after childhood.
- **Parapharyngeal space:** beside the pharynx, an inverted pyramid from the skull base to the hyoid; its post-styloid part contains the carotid vessels, internal jugular vein, the last four cranial nerves and the sympathetic chain.

Their infections are described with deep neck space infection.

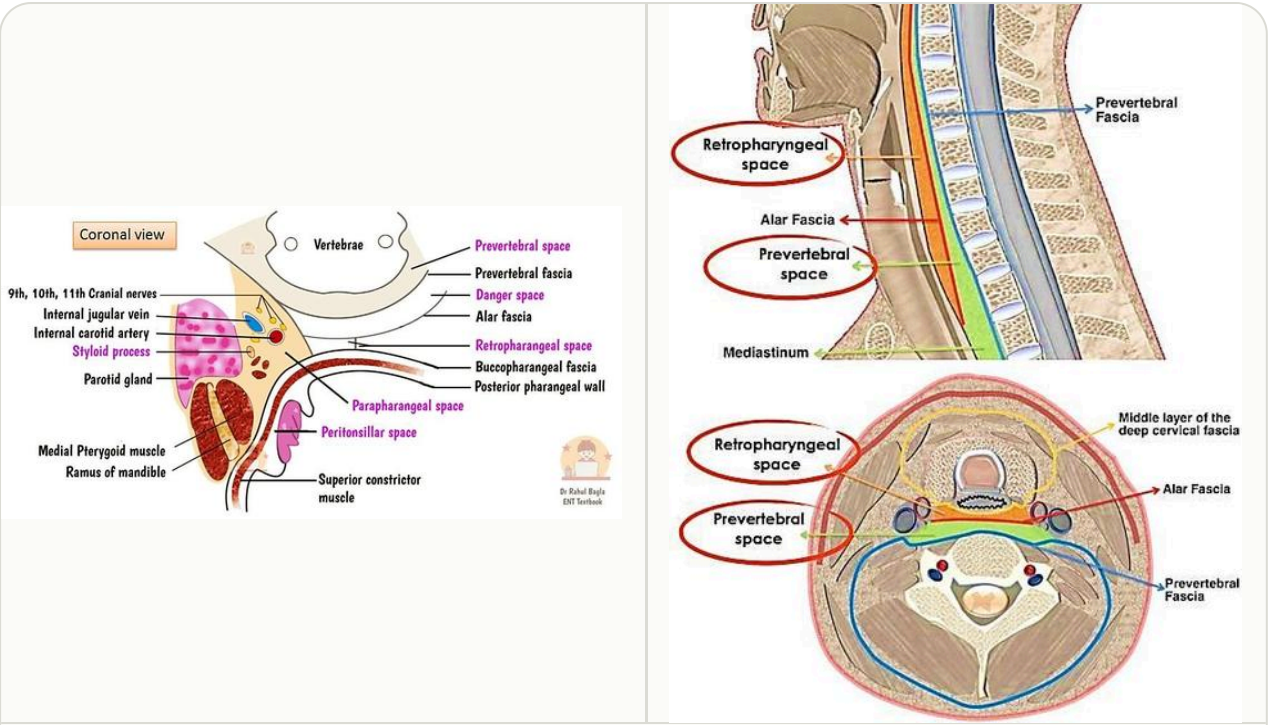


FIGURE 17 The parapharyngeal and retropharyngeal spaces, and the retropharyngeal, danger and prevertebral spaces.

Anatomy atlas · ENT Steps p. 198, ENT Steps p. 277

8 CHAPTER EIGHT Swallowing

Three phases.

8.1 Oral phase

Voluntary. Food is chewed, mixed with saliva and formed into a bolus that the tongue pushes back. When the bolus touches the anterior pillars, receptors supplied by the glossopharyngeal and vagus nerves trigger the pharyngeal phase.

8.2 Pharyngeal phase

Involuntary and lasting under a second, while breathing stops:

1. The soft palate rises and meets Passavant's ridge, closing off the nasopharynx.

2. The vocal folds, false folds and arytenoids close the glottis, and the epiglottis tilts down.
3. The larynx and hyoid rise and move forwards under the suprahyoid and thyrohyoid muscles.
4. The tongue base drives back and the constrictors contract from above down.
5. Cricopharyngeus relaxes and is pulled open by the forward movement of the larynx, the most important factor in opening the sphincter.

8.3 Oesophageal phase

A primary peristaltic wave carries the bolus to the stomach in three to six seconds, and the lower oesophageal sphincter relaxes.

SWALLOWING NERVES

Sensation by V, IX and X; muscles by V (jaw, mylohyoid, tensor), VII (lips, face), IX (stylopharyngeus), X (palate, pharynx, larynx, cricopharyngeus) and XII (tongue). Incoordination between pharyngeal contraction and cricopharyngeal relaxation is thought to underlie the pharyngeal pouch.

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Layers of the pharyngeal wall** and the sinus of Morgagni.
- **Killian's dehiscence.**
- **Waldeyer's ring.**
- **Boundaries of the nasopharynx** and the fossa of Rosenmüller.
- **Subsites of the hypopharynx** and the internal laryngeal nerve.
- **Nerve supply of the pharynx** and referred otalgia.

- **Phases of swallowing.**

9.2 Common pitfalls

COMMON PITFALLS

- Forgetting that stylopharyngeus (IX) and tensor veli palatini (V3) are the exceptions to vagal supply.
- Placing the valleculae in the hypopharynx; they belong to the oropharynx.
- Looking only at the ear in a patient with otalgia and a normal drum.

Written for this app from Oxford Specialist Handbook ch. 6, Dhingra ch. 47 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

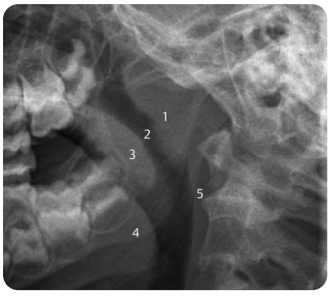
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions

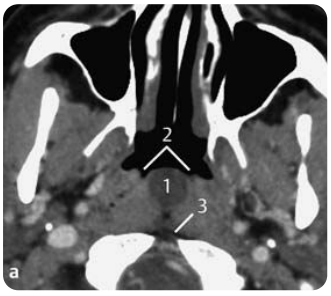


Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

PHARYNX, TONSILS & OESOPHAGUS · CONGENITAL

Thornwaldt Cyst



CT, axial. During endoscopy, a rounded expansion on the posterior wall of the nasopharynx in a patient with sinonasal symptoms was noted. CT…